Chess Game — Phase 2 GUI Design

Course: CS 3354 | **Implementation:** Java 21 + Swing

Team: Mohammad Ishar and teammate (add teammate full name before submitting).

Repository: Add the verified shared GitHub repository URL in Canvas. Phase 2 code is isolated in `ChessGame_Phase2/`.

Purpose. Phase 2 adds an 8×8 GUI, two colors of chess pieces in standard starting squares, alternating player turns, click-to-move and drag-and-drop, captures, and a winner popup after a king capture. In accordance with the instructions, full chess move validation is intentionally not required.

1. Starting board



Figure 1. Starting GUI: 8×8 alternating squares, all 32 chess pieces, and the move-history/capture panel.

Implementation note: Phase 2 reuses the Phase 1 piece classes while keeping the GUI and simplified Phase 2 move logic separate. White begins the game.

2. Moving and capturing pieces

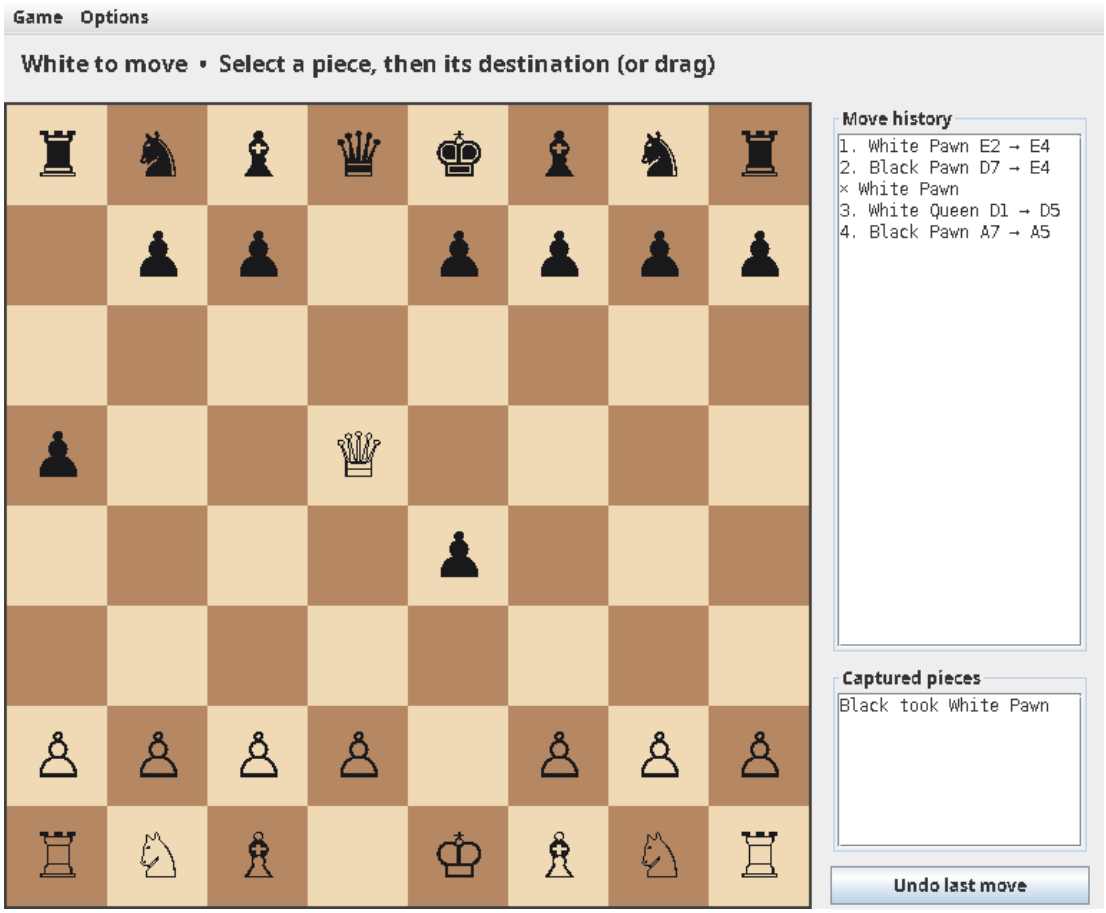


Figure 2. After White E2–E4, Black D7–E4 (capture), White D1–D5 and Black A7–A5. The captured pawn disappears, and the sidebar records the moves and capture.

3. Winner notification



Figure 3. White moves its queen from D5 to E8 to capture the Black King. A modal dialog announces “White wins!”; the application exits when the user confirms the popup.

4. Extra GUI feature 1 — Game controls



Figure 4. Game menu includes New Game, Save Game, and Load Game. Saves use a local text file containing the full board, turn, and move history.

5. Extra GUI feature 2 — Settings



Figure 5. Settings allows selection of Classic/Gray/Green squares, Unicode chess symbols or piece letters, and Small/Medium/Large board sizes. Click OK to apply changes immediately.

6. Extra GUI feature 3 — Game history and Undo

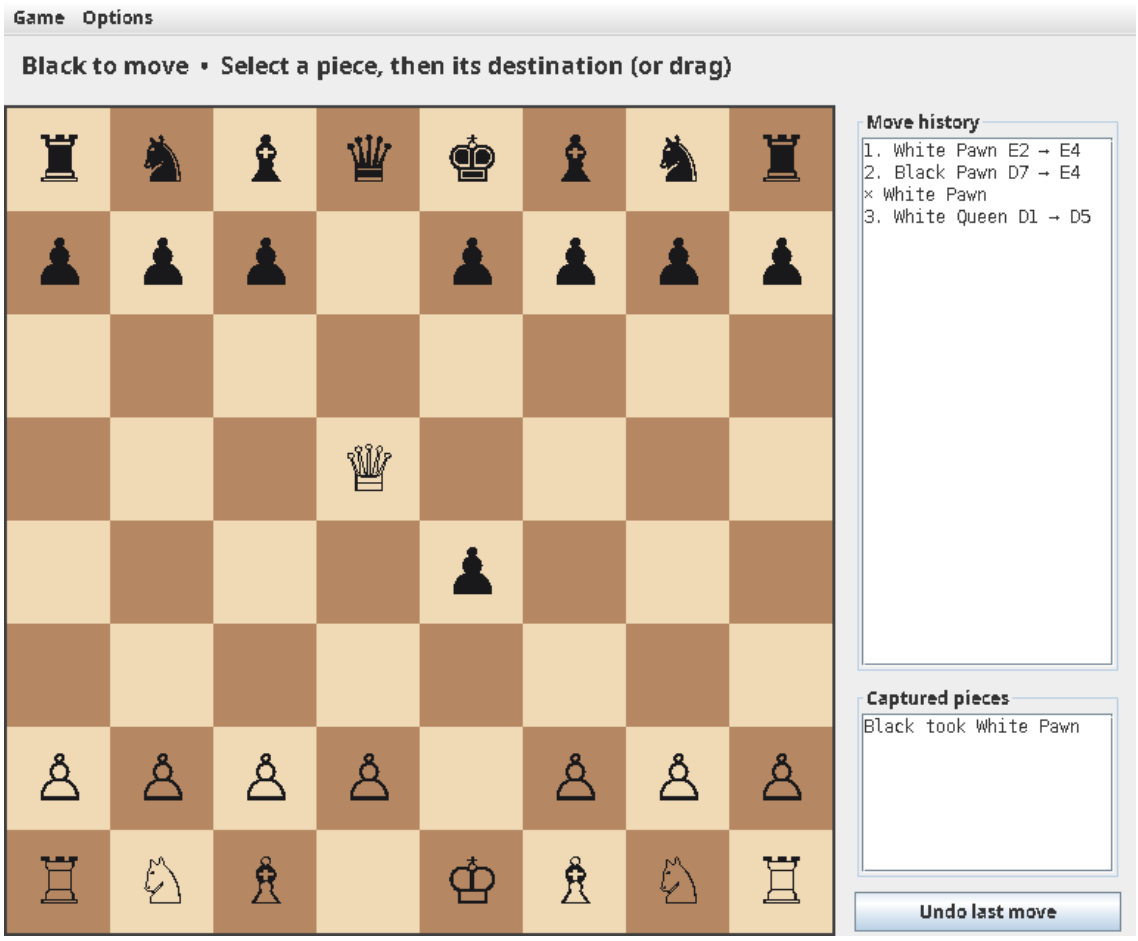


Figure 6. The move-history panel tracks each move and captured pieces. Undo last move restores the previous square, the active player, and any captured piece.

7. Verification and comments

The Java source compiled successfully with `javac`. Automated model checks passed for the opening layout, turn changes, captures, Undo, New Game, save/load round trips, undo after loading, king capture, and invalid save files. GUI screenshots above were recorded from the running Swing application in a virtual desktop test environment.

Before submitting: Both teammates should run the game on their own computers, manually verify real mouse drag-and-drop and menu actions, confirm the final names, and upload the updated Phase 2 folder to their shared GitHub repository. These screenshots demonstrate the test run; they are not screenshots taken on a teammate's computer.

Known scope: The interface intentionally omits check/checkmate and other full chess rules because the Phase 2 instruction explicitly does not require movement validation. The original Phase 1 console project remains unchanged.